

CROSS-PROJECT INTERVIEW — 20 Questions

Comparing All Three Projects & Internship

CQ1. Which of your three projects are you most confident about technically?

Answer:

- I'm most confident about the **Agentic AI Customer Support System** because I actively implemented it during my internship, worked with experienced engineers, and the technologies (LangChain, LangGraph, RAG) are something I used daily.
 - For the **AIxAI Negotiation System**, I designed and implemented it myself, so I understand every decision, but it was a solo/team project without professional mentorship.
 - For the **Geo-Fenced Firearm System**, I'm confident about the embedded systems and RFID logic, but the hardware debugging made it more unpredictable than purely software projects.
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CQ2. Which project had the hardest technical problem?

Answer:

- The **Geo-Fenced Firearm System** had the hardest technical problems because it combined hardware and software — bugs could be in the Embedded C code, the hardware wiring, the RFID module's communication protocol, or the power supply.
 - Debugging embedded systems without a traditional debugger is fundamentally harder than debugging software — you rely on serial output, LED indicators, and logic analyzers.
 - In the other two projects, Python has a rich debugging ecosystem. In Embedded C on ESP32, debugging is significantly more constrained.
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CQ3. You used Python in both the AIxAI project and the internship. Why not Java, which is also on your resume?

Answer:

- Python was the right tool for both because:

- Both involve data processing (NumPy, Pandas) and AI/ML components — Python's ecosystem is far superior to Java's for these tasks.
 - FastAPI and Flask (Python) are faster to prototype with than Spring Boot (Java).
 - LangChain and LangGraph only have Python SDKs (and partially JavaScript) — no Java equivalent.
 - Java would have been a better choice if the projects required high-performance concurrent processing, Android app development, or enterprise Spring-based systems.
 - I would use Java for Android development, high-throughput microservices, or systems where JVM performance characteristics are important.
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CQ4. What did you learn from the internship that you can apply to your own projects?

Answer:

- **Production thinking:** At the internship, I saw how systems need to handle edge cases, failures, and monitoring — not just the happy path. I've applied this thinking to how I design failure handling in my projects.
 - **LangChain and LangGraph patterns:** Before the internship, I knew about AI conceptually. The internship gave me hands-on experience connecting LLM components in a real workflow.
 - **Code organization at scale:** Working on a team codebase taught me about module separation, naming conventions, and writing code that others can read.
 - **Evaluation mindset:** At the internship, we tracked retrieval quality metrics. I now think about how to measure whether a system is actually working well, not just "does it produce output."
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CQ5. Which of your projects is the most scalable as designed?

Answer:

- The **Agentic AI Customer Support System** is the most naturally scalable because:
- The core components (RAG service, intent classifier, agentic workflow) are stateless microservices that can be scaled horizontally.
- LLM API calls are inherently scalable — just add more concurrent requests.

- ChromaDB can be replaced with a managed vector database (Pinecone) as scale grows.
 - The **AIxAI Negotiation System** is moderately scalable — the negotiation engine can be made stateless and scaled horizontally.
 - The **Geo-Fenced Firearm System** requires careful infrastructure design for scaling — each additional reader needs network connectivity, device management, and secure provisioning.
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CQ6. Which project has the biggest security risk?

Answer:

- The **Geo-Fenced Firearm Safety System** has the highest stakes — a security compromise could allow unauthorized weapon activation or unauthorized weapon locking, which has physical safety implications.
 - Specifically: RFID tag cloning, rogue reader attacks, and firmware tampering are all risks in a safety-critical system.
 - The **Agentic AI System** has a significant risk of prompt injection — malicious users manipulating the AI into performing unauthorized actions.
 - The **AIxAI Negotiation System** has financial risks — compromised business constraints could lead to unprofitable offers.
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CQ7. If you had to pick ONE project to deploy in production today, which would it be?

Answer:

- The **Agentic AI Customer Support System** is closest to production-ready because:
 - It was built in an internship environment with professional guidance.
 - The technology stack (Python, FastAPI, LangChain, LangGraph, ChromaDB) is well-suited for production.
 - Software systems are easier to update, patch, and deploy than hardware.
 - The RAG architecture is a proven production pattern.

- The Firearm System would require significant hardware ruggedization, security hardening, and regulatory approval before production.
 - The AIxAI System would need real data integration and proper ML model evaluation before production.
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CQ8. Which project demonstrates your strongest backend knowledge?

Answer:

- The **Agentic AI Customer Support System** — because it involves:
 - API design with FastAPI
 - RAG pipeline with vector database integration
 - Agentic workflow orchestration with LangGraph
 - Tool integration with external APIs
 - Error handling across multiple system components
 - This project has the broadest backend coverage — data pipelines, API development, external integrations, and AI system design.
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CQ9. Which project demonstrates your strongest AI knowledge?

Answer:

- The **Agentic AI Customer Support System** from the internship demonstrates the deepest AI knowledge:
 - RAG implementation with semantic search
 - Embedding models and vector databases
 - LLM prompt engineering
 - Agentic AI with tool selection and execution
 - LangChain and LangGraph (modern AI engineering frameworks)
- The **AIxAI Negotiation System** demonstrates AI knowledge in a different dimension — classical AI (utility-based agents, decision theory) and ML (churn prediction).
- Together they show breadth: classical AI decision-making AND modern generative AI/RAG.

CQ10. Which project would you completely rebuild if you started today?

Answer:

- I would rebuild the **AIxAI Negotiation System** most extensively because:
- The preference weights were statically defined — I'd now build behavioral inference to learn them from actual customer actions.
- I'd add proper ML model training and evaluation for the churn predictor using real data.
- I'd integrate an A/B testing framework to measure actual business impact.
- I'd add an event-driven trigger system instead of a polling scheduler.
- For the Firearm System, I'd add RFID encryption from day one rather than treating it as an optional improvement.
- For the internship system, I'd add evaluation metrics and a reranking step from the start.

CQ11. You have React in your resume but your AI projects use Python backends. How would React connect to your FastAPI backend?

Answer:

- React runs in the browser. FastAPI runs on a server.
- React makes **HTTP API calls** using `fetch()` or `axios` to the FastAPI endpoints.
- Example: When a user types a message in the chat UI:

```
javascript const response = await axios.post('/api/chat', { conversation_id: conversationId, message: userMessage }); setResponse(response.data.response);
```
- For real-time streaming responses, React can use `EventSource` (Server-Sent Events) or `WebSockets`.
- For the Firearm dashboard, React polls `GET /api/events` every few seconds or uses `WebSocket` to receive real-time detection events.

CQ12. You listed Docker on your resume. How would you containerize any one of your projects?

Answer:

For the Agentic AI system:

```
# Dockerfile for FastAPI app
FROM python:3.11-slim

WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

# Don't embed secrets in image - use environment variables
ENV OPENAI_API_KEY=""
ENV CHROMA_HOST="chromadb"

EXPOSE 8000
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

- **Best practices I know:**

- Use slim base images to reduce image size.
- Copy requirements.txt before the app code (leverages Docker layer caching — only reinstalls dependencies when requirements change).
- Never put secrets in the Dockerfile — use environment variables or secrets managers.
- Use .dockerignore to exclude __pycache__, .env, .git from the image.

CQ13. You listed AWS. How would you deploy your Agentic AI customer support system on AWS?

Answer:

- **ECS (Elastic Container Service):** Run the FastAPI containers. Auto Scaling adds instances during traffic spikes.
- **ALB (Application Load Balancer):** Distributes traffic across ECS tasks.
- **RDS (PostgreSQL):** Managed database for conversation history and logs.

- **ElastiCache (Redis):** Cache LLM responses for duplicate queries, store session state.
 - **Secrets Manager:** Store OpenAI API key, database credentials securely.
 - **CloudWatch:** Monitor API latency, error rates, LLM token usage.
 - **S3:** Store raw company documents before ingestion.
 - **Lambda:** Run the document ingestion pipeline triggered when new documents are uploaded to S3.
 - **Route 53 + ACM:** DNS and SSL certificate management.
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CQ14. Looking at all three projects, where did DSA knowledge actually matter?

Answer:

- **AlxAI Negotiation System:** The customer preference lookup (dict) is $O(1)$ — hash table. The negotiation loop has $O(\text{rounds})$ time complexity. The churn feature computation involves sorting and filtering arrays.
 - **Geo-Fenced Firearm System:** The authorized ID lookup uses an array — $O(n)$ linear search. Improving to a sorted array with binary search would be $O(\log n)$. For very large authorized lists, a hash set would be $O(1)$.
 - **Agentic AI System:** Vector similarity search is essentially a nearest-neighbor search problem — a fundamental DSA/algorithm problem optimized using ANN (Approximate Nearest Neighbor) with HNSW graphs.
 - **OOP:** Applied in the agent classes (CustomerAgent, BusinessAgent), LangGraph node functions, and tool definitions.
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CQ15. What is the biggest weakness in your technical resume?

Answer:

- I'll be honest: **my projects are primarily prototypes and simulations, not production-deployed systems.**
- I haven't yet managed the full lifecycle of a system in production — monitoring, incident response, on-call debugging of live issues.
- My embedded systems experience is limited to the firearm prototype — I don't have deep industry embedded systems experience.

- I'm actively addressing this by studying system design, following production engineering blogs, and building more robust versions of my projects.
 - My strength is that I can understand and connect multiple layers — hardware, backend, AI — which gives me a broad foundation to build on.
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CQ16. Which technology across all your projects would you most want to replace with something better today?

Answer:

- In the **Firearm System**: Replace basic RFID UID matching with **encrypted challenge-response RFID authentication** (e.g., using ISO 15693 with cryptographic challenges). The current approach is vulnerable to tag cloning.
 - In the **AlxAI System**: Replace the static preference weights with a **behavioral preference inference model** that learns from customer actions.
 - In the **Agentic AI System**: Add a **retrieval reranker** (like Cohere Rerank) on top of ChromaDB's initial vector search for better precision.
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CQ17. If an interviewer asks you "which project taught you the most," what do you say?

Answer:

- The **Agentic AI internship** taught me the most because:
 - I worked with real engineers on a real product.
 - I was exposed to production engineering practices — code reviews, version control discipline, error handling, monitoring.
 - I learned a completely new technology stack (LangChain, LangGraph, vector databases) in a professional environment.
 - I received feedback that improved my code quality and system thinking.
 - My own projects taught me **ownership and accountability** — when it breaks, there's no one else to fix it. That built deeper debugging skills and forced me to understand every component fully.
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CQ18. How do your CS fundamentals (DBMS, OOP, OS, DSA) connect to your projects?

Answer:

- **DBMS:** Schema design for all three projects, indexing decisions, normalization, transaction handling for concurrent updates.
 - **OOP:** CustomerAgent and BusinessAgent classes (AIxAI), LangGraph node function organization, React component architecture.
 - **OS:** Process management for the ESP32 (watchdog timer, interrupt handling), MQTT connection management, understanding of how the server OS handles concurrent HTTP requests.
 - **DSA:** ID lookup data structures (hash sets for $O(1)$), ANN algorithms for vector search, negotiation loop complexity analysis, sorting and filtering in data preprocessing.
 - **Networking:** HTTP REST APIs, MQTT protocol for IoT, TLS security, WebSockets for real-time dashboard updates.
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CQ19. A recruiter asks: "Why should we hire you over someone who only specializes in backend or AI?" What do you say?

Answer:

- My unique value is **breadth without sacrificing depth in the areas I've focused on**.
 - I can work across the full stack — embedded hardware → backend API → AI pipeline → frontend dashboard. I understand how all layers connect.
 - This makes me effective in systems where the AI team, backend team, and frontend team need to communicate — I can bridge those gaps.
 - I've applied backend concepts (FastAPI, databases, APIs) in an AI context and embedded systems context — not just in isolation.
 - I'm not claiming to be a specialist in every area — I'm claiming to be someone who can navigate across them and understand the system as a whole.
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CQ20. Final question: What do you want to learn next?

Answer:

- **MLOps and production ML:** How to train, evaluate, deploy, and monitor ML models in production — model versioning, data drift detection, retraining pipelines.
 - **Distributed systems:** Deeper understanding of distributed databases, consensus algorithms, and fault tolerance — because all my production scale-up answers require these concepts.
 - **System design at scale:** I understand the concepts but want to practice designing real high-scale systems — preparing for senior-level system design interviews.
 - **Cloud architecture:** Getting hands-on with AWS or GCP deployments — moving beyond theoretical architecture to actually deploying and operating a system in the cloud.
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RAPID-FIRE TECHNICAL ROUND — 30 Questions

RF1. What is the difference between a process and a thread?

- A **process** is an independent program with its own memory space.
 - A **thread** is a unit of execution within a process — multiple threads share the same memory.
 - Threads are lighter than processes but share state, so require synchronization.
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RF2. What is a deadlock?

- When two or more processes are each waiting for the other to release a resource — neither can proceed.
 - Conditions: mutual exclusion, hold and wait, no preemption, circular wait. Remove any one to prevent deadlock.
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RF3. What is the difference between stack and heap memory?

- **Stack:** Function call frames, local variables. LIFO structure. Fast, automatically managed. Limited size.
 - **Heap:** Dynamic memory allocation (`new`, `malloc`). Slower, must be manually freed (or garbage collected). Large size.
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RF4. What is polymorphism in OOP?

- The ability for the same method name to behave differently based on the object.
 - **Compile-time (overloading)**: Same method name, different parameters.
 - **Runtime (overriding)**: Subclass provides a different implementation of a parent class method.
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RF5. What is the difference between `==` and `.equals()` in Java?

- `==` compares **references** (memory addresses for objects).
 - `.equals()` compares **values** (content of the objects).
 - For Strings in Java: always use `.equals()` for value comparison.
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RF6. What is normalization in databases?

- The process of organizing a database to reduce redundancy and improve data integrity.
 - **1NF**: Atomic values, no repeating groups.
 - **2NF**: 1NF + no partial dependencies on composite keys.
 - **3NF**: 2NF + no transitive dependencies.
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RF7. What is an index in a database and what is its trade-off?

- An index speeds up data retrieval by creating a searchable data structure (B-tree or hash).
 - Trade-off: Faster reads, slower writes (every write must also update the index), more storage.
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RF8. What is the difference between SQL `JOIN` types?

- **INNER JOIN**: Returns rows with matching values in both tables.
- **LEFT JOIN**: Returns all rows from the left table + matching from right. Nulls for no match on right.

- **RIGHT JOIN:** Opposite of LEFT JOIN.
 - **FULL OUTER JOIN:** All rows from both tables. Nulls where no match.
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RF9. What is a REST API?

- Representational State Transfer — an architectural style for designing networked APIs.
 - Stateless: each request contains all information needed.
 - Uses standard HTTP methods: GET, POST, PUT, PATCH, DELETE.
 - Resources identified by URLs. Responses typically in JSON.
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RF10. What is the difference between GET and POST?

- **GET:** Retrieve data. Parameters in URL. Idempotent (safe to repeat). Cached by browsers.
 - **POST:** Submit data. Parameters in request body. Not idempotent. Not cached.
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RF11. What is HTTP status code 404, 401, 403, 500, 429?

- **404:** Not Found — resource doesn't exist.
 - **401:** Unauthorized — authentication required.
 - **403:** Forbidden — authenticated but not authorized.
 - **500:** Internal Server Error — server-side bug.
 - **429:** Too Many Requests — rate limit exceeded.
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RF12. What is async/await in Python?

- A way to write asynchronous code that doesn't block while waiting for I/O operations.
 - `async def` defines a coroutine. `await` suspends it until the awaited operation completes.
 - While one coroutine awaits, another can run — enables concurrency without threads.
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RF13. What is FastAPI's `async def` vs `def` endpoint?

- `async def`: Non-blocking. FastAPI runs it directly in the event loop. Use when the function does I/O-bound work (DB queries, API calls) that can be awaited.
 - `def`: Blocking. FastAPI runs it in a thread pool to prevent blocking the event loop.
 - Mixing correctly is important — calling a blocking function in `async def` without `await` blocks the event loop.
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RF14. What is a vector embedding?

- A numerical representation of data (text, image, audio) as a dense vector (list of floats).
 - Captures semantic meaning — similar inputs have similar (close) vectors in the embedding space.
 - Generated by a trained neural network (embedding model).
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RF15. What is cosine similarity?

- A measure of similarity between two vectors based on the angle between them.
 - Value range: -1 (opposite) to 1 (identical).
 - For embeddings: higher cosine similarity = more semantically similar text.
 - Formula: $\cos(\theta) = (A \cdot B) / (|A| \times |B|)$ (dot product divided by product of magnitudes).
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RF16. What is the difference between RAG and fine-tuning?

- **RAG**: Retrieves relevant documents at query time and injects as context. Knowledge is external. Easy to update.
 - **Fine-tuning**: Trains the model's weights on new data. Knowledge is internal. Expensive to update.
 - RAG for frequently changing, company-specific data. Fine-tuning for changing style/capability/general knowledge.
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RF17. What is LangChain?

- A Python framework for building applications with LLMs.
 - Provides abstractions for: prompts, chains, retrievers, tools, memory, agents.
 - Makes it easier to connect LLMs with vector databases, external APIs, and data sources.
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RF18. What is LangGraph?

- A library built on LangChain for creating stateful, multi-step agentic workflows as directed graphs.
 - Nodes = processing steps. Edges = transitions (including conditional). State = shared data across nodes.
 - Supports cycles (loops) unlike linear chains.
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RF19. What is the difference between LangChain Agents and LangGraph?

- **LangChain Agents (ReAct):** LLM autonomously decides which tool to call next at each step. More autonomous, less predictable.
 - **LangGraph:** Developer defines the workflow graph explicitly. LLM may be used at nodes but the overall flow is controlled. More predictable and auditable.
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RF20. What is React's virtual DOM?

- React keeps a lightweight in-memory representation of the real DOM (the virtual DOM).
 - When state changes, React first updates the virtual DOM, then diffs it with the previous virtual DOM.
 - Only the changed parts are updated in the real DOM — minimizing expensive real DOM operations.
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RF21. What is the difference between `useState` and `useEffect` in React?

- **useState:** Declares a state variable in a functional component. Re-renders the component when state changes.
 - **useEffect:** Runs a side effect after render (e.g., fetch data from API, subscribe to events). Can clean up when component unmounts.
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RF22. What is Flutter and how is it different from React Native?

- **Flutter:** Google's UI framework using Dart language. Compiles to native code. Renders its own widgets — looks the same on all platforms.
 - **React Native:** Facebook's framework using JavaScript. Uses native platform components — looks native on each platform.
 - Flutter: more consistent UI, better performance for complex UIs. React Native: leverages existing JS/React knowledge.
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RF23. What is Docker and what problem does it solve?

- Docker is a containerization platform — it packages an application and all its dependencies into a container.
 - Problem solved: "Works on my machine but not on production." A container runs identically everywhere because the environment is packaged with the code.
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RF24. What is the difference between a Docker container and a virtual machine?

- **VM:** Virtualizes an entire OS on top of the hypervisor. Heavy, slow to start, large.
 - **Container:** Virtualizes only the application and its dependencies. Shares the host OS kernel. Lightweight, fast to start.
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RF25. What is Git rebase vs merge?

- **Merge:** Combines two branches, creates a merge commit, preserves full history.
- **Rebase:** Moves the commits of one branch onto the tip of another, creating a linear history. No merge commit.
- Use merge for preserving history. Use rebase for cleaner, linear history.

RF26. What is the difference between SQL and NoSQL?

- **SQL:** Structured, relational, fixed schema, ACID transactions, good for complex queries and relationships.
- **NoSQL:** Flexible schema (document, key-value, graph, column-family), horizontal scaling, good for unstructured or high-volume data.

RF27. What is caching and why is it used?

- Caching stores frequently accessed data in a fast-access location (like memory) instead of recomputing or re-fetching from a slow source (like a database).
- Reduces latency, reduces database load, improves throughput.
- Common tools: Redis, Memcached. Common strategies: write-through, write-back, eviction policies (LRU, LFU).

RF28. What is a message queue? Give an example use case.

- A message queue decouples producers and consumers. The producer sends a message to the queue without knowing who will process it. The consumer reads from the queue independently.
- Use case: An order placed on an e-commerce site → the "place order" service sends a message to a queue. Separately, the "send confirmation email" service and "update inventory" service each consume the message independently.
- Tools: RabbitMQ, Apache Kafka, AWS SQS.

RF29. What is the time complexity of binary search?

- $O(\log n)$ — each comparison halves the search space.
- Requires a sorted array.
- Much better than linear search $O(n)$ for large datasets.

RF30. What is ACID in databases?

- **Atomicity:** A transaction is all-or-nothing. Either all operations succeed or none do.
 - **Consistency:** A transaction brings the database from one valid state to another. Rules (constraints, triggers) are enforced.
 - **Isolation:** Concurrent transactions don't interfere with each other.
 - **Durability:** Once a transaction is committed, it persists even if the system crashes.
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FINAL PREPARATION NOTES

How to Use This Document

- **Daily review:** Go through the "Most Important 10" for each project every day.
- **Practice out loud:** Don't just read — speak the answers. Record yourself and listen back.
- **Know your "Never Get Wrong" questions cold** — these are the ones that will determine whether the interviewer believes you built the project.
- **Use the 5-Minute Project Explanation** as your opening pitch for each project when asked "Tell me about this project."
- **For rapid-fire:** Cover all 30 in one session. These should take no more than 30 seconds each.

Interview Mindset

- You built these projects. Speak with ownership, not hesitation.
- When asked about unspecified details, say: *"That specific detail was not tracked in my implementation notes — I'd need to verify. But the approach I would use is..."* — and give the correct technical answer.
- When challenged, don't immediately back down. If you're confident, defend your decision. If the interviewer has a valid point, acknowledge it: *"That's a fair point — a better approach would be X."*
- Honesty builds more trust than bluffing. If you don't know something, say: *"I'm not sure about that specific detail, but here's how I'd approach figuring it out."*